Here are the questions and their corresponding answers for the Introduction to Statistics & Data Collection section:

📊 Introduction to Statistics & Data Collection

Questions & Answers

1. An e-commerce firm wants to understand customer satisfaction using a 1-5 rating scale. What type of data is this and why?

- **Answer:** This is **Ordinal Data**.
- **Reason:** It is **categorical** data where the ratings (1, 2, 3, 4, 5) have a logical **order** (5 is better than 1), but the difference between the levels (e.g., the perceived gap between a rating of 4 and 5) is subjective and cannot be precisely measured or assumed to be equal.

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2. A political analyst conducts a survey in 5 districts and chooses 100 people from each based on their occupation. What sampling method is this?

- **Answer:** This is **Stratified Sampling**.
- **Reason:** The entire population (voters) is first divided into non-overlapping subgroups called **strata** (the 5 districts, or groups based on occupation). Then, a fixed number of samples (100 people) is drawn from *each* stratum.

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3. A fitness app collects users' average daily steps from Android phones. What type of data collection is this, and what bias could it introduce?

- **Answer:**
    - **Type of Data Collection: Observational Data Collection**.
    - **Bias:** It introduces **Selection Bias**, specifically **Under coverage Bias**.
- **Reason:** This method of data collection only observes data generated by a specific group (**Android phone users**). It systematically excludes individuals who use the app on other platforms (like iOS/iPhone), potentially leading to an inaccurate average for the overall population.

📊 **Measures of Central Tendency**

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4. The average salary in a small IT firm is ₹60,000, but one founder earns ₹10,00,000. What is the best central tendency measure to report to investors?

- **Answer:** The **Median**.
- **Reason:** The single high outlier (₹10,00,000) heavily skews the Mean (average), making it unrepresentative of the typical salary. The Median is robust to these extreme values and reflects the central value of the distribution better.

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5. A college tracks students' grades (A, B, C, D). Which measure(s) of central tendency can be used?

- **Answer:** Only the **Mode**.
- **Reason:** Grades are **Ordinal** (ranked categorical data). Since we cannot calculate meaningful differences or averages, only the Mode (the most frequent grade) is mathematically appropriate.

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6. In a team of 5 people with salaries: ₹30K, ₹30K, ₹40K, ₹50K, ₹8L. Find the mode, and explain its relevance here.

- **Answer:**
  - **Mode: ₹30K**.
  - **Relevance:** The mode highlights the **most common salary level** in the team. Despite the presence of extreme salaries (like ₹8L), the mode tells you the value shared by the largest number of team members.

📊 Measures of Dispersion

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7. Two stores report the same average monthly sales (₹50,000). Store A has std. deviation of ₹2,000 and Store B has ₹10,000. Which store is more consistent?

- **Answer: Store A**.
- **Reason: Consistency** is inversely related to variability (dispersion). Store A has a lower **Standard Deviation** (₹2,000), meaning its monthly sales figures cluster much closer to the average, indicating more predictable and consistent performance.

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8. A delivery company tracks package weights. Why might Mean Absolute Deviation (MAD) be preferred over variance in operational reports?

- **Answer: MAD is easier to interpret operationally.**
- **Reason:** MAD is measured in the **original units** of the data (e.g., kilograms or pounds), which is immediately understandable to staff. Variance is measured in squared units (e.g., kg^2), which is abstract and not intuitive for daily operations.

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9. You're comparing two product prices: Product A (low range, high variation), Product B (higher price, consistent). Which statistic would help?

- **Answer: Coefficient of Variation (CV)**.
- **Reason:** CV expresses the standard deviation as a percentage of the mean (CV) = sigma / mu). This allows for a **standardized comparison of relative variability** between two datasets that have different absolute scales (different prices or means).

📊 **Percentiles, Quartiles, IQR**

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10. A student scored in the 90th percentile in a test. What does this mean?

- **Answer:** It means the student's score was **equal to or better than 90%** of all other scores recorded in that test group.

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11. A dataset has Q1 = 40, Q3 = 70. Identify potential outliers using IQR.

- **Answer:** Potential outliers are any data points **below -5** or **above 115**.
- **Calculation:**
  - IQR = Q3 - Q1 = 70 - 40 = 30.
  - *Lower Bound* = $Q1 - 1.5 \times IQR = 40 - 1.5 \times 30 = -5$.
  - *Upper Bound* = $Q3 + 1.5 \times IQR = 70 + 1.5 \times 30 = 115$.

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12. A company shows employee salaries using a box plot. How would you explain a long upper whisker?

- **Answer:** A long upper whisker indicates that the **top 25% of salaries** (the data between Q3 and the maximum value) are **highly spread out** over a wide range. This often points to a few very high salaries pulling the maximum value far away from the median group.

📊 **Univariate & Bivariate Analysis**

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13. Sales data shows a bell-shaped histogram. What does this suggest about the distribution?

- **Answer:** It suggests the data follows a **Normal (or Gaussian) Distribution**.
- **Reason:** A bell shape indicates that the data is symmetrically distributed around the centre, and most values cluster near the mean.

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14. A scatter plot between study time and marks shows an upward trend. What type of correlation is it?

- **Answer: Positive Correlation**.
- **Reason:** An upward trend means that as one variable (study time) increases, the other variable (marks) also tends to increase.

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15. A retail manager plots customer age vs product category preference. What kind of analysis is this?

- **Answer: Bivariate Analysis**.
- **Reason:** The manager is examining the relationship between **two variables** (customer age and product category) simultaneously.

📊 Probability & Distributions

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16. If the chance of rain tomorrow is 20%, what's the probability it doesn't rain?

- **Answer: 0.80** or **80%**.
- **Reason:** This is found using the complement rule: P(No Rain) = 1 - P(Rain).

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17. You flip a fair coin 4 times. What's the probability of getting exactly 2 heads?

- **Answer: 0.375** (or 3/8).
- **Reason:** This is solved using the Binomial Probability Formula: P(X=2) = (4\2) X(0.5) ^2 X (0.5) ^2 = 6 X0.0625.

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18. What real-life scenario can approximate a binomial distribution?

- **Answer: The number of people who pass a certification exam** out of a class of 50 students.
- **Reason:** This fits the criteria for a binomial distribution: fixed number of trials (50 students), only two possible outcomes per trial (Pass/Fail), and independent trials.

📊 **Hypothesis Testing**

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19. A food company claims their chocolate bar weighs 100g. A test on a sample shows average weight = 98.5g. What is the null and alternate hypothesis?

- **Answer:**
    - **Null Hypothesis (Ho):** mu = 100 g (The true mean weight is 100g).
    - **Alternate Hypothesis (Ha):** mu not equal to 100 g (The true mean weight is not 100g).

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20. What does a p-value of 0.03 mean in a two-tailed test at 5% significance?

- **Answer:** It means you **Reject the Null Hypothesis** (Ho).
- **Reason:** Since the p-value (0.03) is less than the significance level (alpha = 0.05), the observed result (the difference from the claim) is statistically significant.

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21. A retail firm wants to test if sales in two regions differ significantly. Which test should they use?

- **Answer: Two-Sample T-Test** (or Independent Samples T-Test).
- **Reason:** This test is used specifically to compare the means of a quantitative variable (sales revenue) between two independent populations (Region 1 and Region 2).